TECHNICAL EVALUATION — SUBMERSIBLE PUMP

1. Evaluation output

Field	Value
Report	submersible_pump_EN201222034S.pdf (89 pages)
Standards	IEC 60335-1:2010 (Part 1); IEC 60335-2-41:2021 (Part 2 2-41)
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2. Report identity

Field	Value
Report number	EN201222034S
Laboratory	EMTEK(NINGBO) CO., LTD.
Report date	Dec 28, 2025
Product declared	Submersible pump
Declared model	SPC55068
Standards claimed	IEC 60335-1:2010+A1:2013+A2:2016; IEC 60335-2-41:2025 on the test specification; IEC 60335-2-41:2012 in the compliance statement
Standards evaluated	US IEC 60335-1:2010; ILNAS-EN IEC 60335-2-41:2021, incorporating IEC 60335-2-41:2012

3. Summary

Field	Value
Final verdict	REJECT — INSUFFICIENT TECHNICAL EVIDENCE
Reason	The tested model and electrical test basis conflict with the declared pump, and several applicable safety assessments lack supporting evidence or are unjustifiably marked N/A. Product compliance is not determined; no measured limit failure for SPC55068 is established by the supplied evidence.
Recommended action	Obtain a verified replacement report based on the correctly identified SPC55068 and the confirmed applicable standard set, including all necessary test records and assessments.
Conclusion	The report cannot support certification of SPC55068 in its present form. The model, rating and supply discrepancies undermine attribution of the reported results, while missing immersion, earthing, cord and heat-and-fire evidence prevents a complete assessment against the supplied standards.
Findings	21 total — Critical 4, High 16, Medium 1, Low 0

4. Findings

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
Critical	5.2	Relevant type tests are carried out on an appliance representing the product assessed.	The declared model is SPC55068, but the summary identifies 4TSC4.0-96-110/1000 as the model receiving full tests; equivalence is not established.	Identify the tested sample and provide a valid assessment of SPC55068.	PDF pp.1–3	Report: verified p.3 Standard: verified in IEC 60335-1:2010 p.26
Critical	7.1	The appliance is marked with its rated voltage or rated voltage range.	The declared marking is 220–240 V, whereas the clause 7.1 entry records 96 V.	Reconcile the sample identification and rated-voltage marking using verified evidence.	PDF pp.2,3,6	Report: no quote given Standard: verified in IEC 60335-1:2010 p.29
Critical	10.1	Power input is assessed against the marked rated power at rated voltage and normal operation.	The declared rating is 550 W, but the test table uses 1,000 W as the rated basis and records 995.2 W at 72 V.	Test and report the declared model against its verified 550 W rating.	PDF pp.2,3,79	Report: verified p.79 Standard: verified in IEC 60335-1:2010 p.37
Critical	4	The particular part supplements or modifies the applicable general part.	The test specification claims IEC 60335-2-41:2025, while the compliance statement claims 2012; the claimed Part 1 amendments and the 2025 particular edition were not supplied. The supplied European particular standard also references A11:2021, which was not supplied.	Confirm the certification standard set and provide the claimed editions and amendments for assessment.	PDF pp.2,3	Report: verified p.2 Standard: verified in IEC 60335-2-41:2021 p.4
High	5.8.1; 11.5; 13.1; 16.2	Applicable tests use the specified supply and voltage derived from the appliance rating.	The declared rating is 220–240 V, 50 Hz, while the particulars state DC and the heating and leakage tables use 76.3 V; these conditions do not establish the required tests for the declared rating.	Repeat the applicable tests using the verified supply and rated-voltage range of SPC55068.	PDF pp.2–4,79,80	Report: verified p.4 Standard: verified in IEC 60335-1:2010 p.27
High	19.7	The stalled-motor test is performed at rated voltage for the applicable period.	The recorded locked-rotor voltage is 116.6 Vdc, inconsistent with the declared 220–240 V, 50 Hz rating; the test duration is not recorded.	Repeat the stalled-motor test at the verified rated supply and record the duration and results.	PDF pp.2,81	Report: verified p.81 Standard: verified in IEC 60335-1:2010 p.54
High	5.7; 11.7	Pump tests use liquid temperature related to the marked maximum, and the heating test uses liquid at that marked temperature.	The report states only that maximum liquid temperature is below 45 °C; the reproduced marking has no maximum temperature and no measured test-liquid temperature is recorded.	Provide the verified temperature marking and test-liquid records or repeat the heating assessment.	PDF pp.3,6,14,79	Report: verified p.3 Standard: verified in IEC 60335-2-41:2021 p.15
High	7.1; 6.2	A submersible pump over 50 W is marked with maximum operating depth, and its IP number and maximum liquid temperature are marked.	The 550 W pump is reported as IP68, but the reproduced marking does not show IP68, operating depth or maximum liquid temperature; operating depth is recorded as N/A.	Provide the actual markings and verify the applicable marking requirements.	PDF pp.2–4,6,7	Report: verified p.4 Standard: verified in IEC 60335-2-41:2021 p.14
High	7.12.1	Installation instructions state maximum total head for pumps rated above 50 W.	The report passes this item by reference to the marking detail, but the reproduced marking gives no maximum total head and the installation instructions are not supplied.	Provide the installation instructions and record the maximum total head.	PDF pp.3,10	Report: verified p.10 Standard: verified in IEC 60335-2-41:2021 p.14
High	15.1.2	Submersible pumps undergo the specified 24 h immersion at pressure determined by maximum operating depth.	A pass is recorded, but maximum operating depth and actual test pressure are absent; the report also refers to 5 m rather than the supplied standard's 15 m alternative.	Provide the marked depth and actual immersion conditions, and repeat the test if the required pressure was not applied.	PDF pp.4,17	Report: verified p.17 Standard: verified in IEC 60335-2-41:2021 p.16
High	19.101; 19.13	Pumps undergo the specified dry-running sequence and meet the applicable abnormal-operation criteria.	The sequence is marked pass, but its test voltage, head, temperatures and post-test electric-strength results are not documented; the appended abnormal-temperature table identifies only 19.7.	Provide the complete 19.101 test record and associated 19.13 results.	PDF pp.24,25,81,82	Report: verified p.81 Standard: verified in IEC 60335-2-41:2021 p.17

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	24.1.4	Applicable thermal motor protectors are assessed with their motor under Annex D conditions.	The test-item particulars indicate a self-resetting thermal cut-out, but the automatic-control and Annex D entries are N/A without identifying the device or establishing whether it is needed for compliance.	Identify the cut-out and document its compliance assessment, including Annex D testing where required.	PDF pp.4,38,62,82	Report: verified p.38 Standard: verified in IEC 60335-1:2010 p.77
High	24.5; 19.7	Auxiliary-winding capacitors are assessed for their ratings and applicable locked-rotor fault conditions.	Clause 24.5 records passes for a motor capacitor, while the capacitor fault tests under 19.7 are N/A and no capacitor is identified in the component table.	Establish whether an auxiliary-winding capacitor is fitted and provide its ratings and applicable fault-test results.	PDF pp.20, 40,82	Report: verified p.40 Standard: verified in IEC 60335-1:2010 p.54
High	25.1; 25.3	A class I submersible pump is provided with a flexible supply cord fitted with a plug.	The pump is identified as class I and submersible, but the pump-specific cord and plug entries are N/A despite a general plug entry marked pass.	Verify the supplied cord and plug and correct the pump-specific assessment.	PDF pp.4,40,41	Report: verified p.40 Standard: verified in IEC 60335-2-41:2021 p.19
High	25.7; 25.8	An outdoor class I submersible pump has the specified cord type and minimum cord length.	Outdoor use is selected in the particulars, but cord type, conductor area and outdoor cord length are not recorded and the relevant entries are N/A.	Identify and measure the supply cord and assess its type, conductor area and length against the applicable requirements.	PDF pp.4,41–43	Report: verified p.42 Standard: verified in IEC 60335-2-41:2021 p.19
High	5.101; 25.14	Pumps are tested as portable unless fixed, and applicable portable pumps undergo the cord-flexing test.	Neither portable nor fixed classification is selected, and the portable-pump flexing test is N/A without a documented fixed-appliance basis.	Establish the pump classification and perform the cord-flexing test if applicable.	PDF pp.4,6,43	Report: verified p.43 Standard: verified in IEC 60335-2-41:2021 p.13
High	25.10; 27.3	A class I supply cord has a correctly connected green/yellow earthing core, and the earthing conductor remains slack longer than current-carrying conductors.	The report identifies a class I appliance with a supply cord, but both checks are N/A without justification.	Inspect and record the earthing-core connections and conductor arrangement.	PDF pp.4,43,49	Report: verified p.43 Standard: verified in IEC 60335-1:2010 p.90
High	27.5	Earthing continuity is tested at the specified current and the calculated resistance does not exceed 0.1 Ω.	A pass is recorded, but no measured resistance, test current or accessible parts tested are provided.	Record the test current and measured resistance for each applicable earthed part.	PDF pp.48,49	Report: verified p.49 Standard: verified in IEC 60335-1:2010 p.91
High	29.1; 29.2	Clearance and creepage assessments use the applicable rated impulse voltage, working voltage, pollution degree and material group.	The tables assess 500 V impulse and ≤50 V working voltage despite the declared 220–240 V rating; basic creepage is only recorded as greater than 2.0 mm, with pollution degree 3 selected and no material group identified.	Reassess each insulation path using verified voltages, material group and measured distances.	PDF pp.2,54,83	Report: verified p.83 Standard: verified in IEC 60335-1:2010 p.97
High	30.1; 30.2; 30.2.1; 30.2.3	Relevant non-metallic parts are assessed for heat and fire resistance using the applicable pump provisions.	Clause 30 and its tests are N/A and the appended table is blank, although the component list includes an insulating bushing; the basis for excluding relevant parts is not documented.	Identify relevant non-metallic parts and provide applicable test results or substantiated exemptions.	PDF pp.55–58,82,87	Report: verified p.55 Standard: verified in IEC 60335-1:2010 p.102
Medium	29.2	Report attribution must be reliable for the evidence to support the assessment.	PDF page 86 carries report number EN200622010S rather than EN201222034S, casting doubt on attribution of the creepage-table continuation.	Obtain a verified replacement page and confirm the integrity of the complete report.	PDF pp.2,86	Report: verified p.86 Standard: requirement not in supplied standards